BRODY ERLANDSON

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EDUCATION

PhD in Statistics, Colorado State University, GPA: 3.87/4.00 — Aug 2022 - Current
M.S. in Data Science, University of Michigan, GPA: 3.97/4.00 — April 2022
B.S. in Mathematics, Eastern Michigan University, GPA: 3.95/4.00 — December 2019
Minor in Philosophy, **Honors**: Deans list and Summa Cum Laude

EXPERIENCE

Algorithms Engineer Intern — May - August 2023 & May - Dec 2024
KLA — *Ann Arbor, MI*

Graduate Teaching Assistant — August 2022 - Present
Colorado State University — *Fort Collins, CO*

Student Research Assistant II: Nielsen Consumer Panel Research — March 2021 - August 2022
University of Michigan — *Ann Arbor, MI*

Lecturer (Part-time) — May 2022 - August 2022
Washtenaw Community College — *Ann Arbor, MI*

Graduate Student Instructor — August 2021 - April 2022
University of Michigan — *Ann Arbor, MI*

Volunteer Graduate Student Peer Mentor at UofM and SOARS at CSU

PROJECTS, RESEARCH, & TEACHING

A Bayesian Functional Concurrent Zero-Inflated Dirichlet-Multinomial Regression Model with Application to Infant Microbiome First Author Paper

- Developed a regression model, FunC-ZIDM, to model time-varying relations between observed covariates and microbial taxa while accounting for zero-inflation, compositionality, and repeated measures.

Assessing Uncertainty and Methods of Evaluation in Deep Learning Classification Models KLA internship project

- Researched and modified state of the art uncertainty quantification and evaluation metrics for deep learning classification techniques for KLA data. The methods were utilized to assess the performance of models on new data. The method was implemented into the codebase.

Deep Learning Classification with Noisy Labels KLA internship project

- Researched and modified state of the art noisy-label deep learning classification techniques for KLA data. The methods utilized semi- and self-supervised learning, alongside advance CNNs. Achieved accuracies nearing those of a clean dataset (within 3-5%) in datasets with up to 20% noise.

Research Interest Bayesian Modeling, Probabilistic Machine Learning, and Causal Inference.
Highlighted Teaching *CSU*: STAA 578 Machine Learning, STAA 575 Applied Bayesian Statistics, STAA 567 Computational and Simulation Methods, *WCC*: MATH 197 Linear Algebra, *UofM*: STATS 413 Linear Regression

SKILLS

Modeling	Regression (Linear to Splines), Supervised and Unsupervised Learning (ML, Deep Learning, and Clustering), and Bayesian Modeling.
Programming Languages	C++, Python, R, and SQL.
Other	Data Manipulation, Git/GitHub, LaTeX, High Performance Computing, and Linux Command-line
Soft Skills	Communication, Problem Solving, Creativity, Project Management, Leadership.